

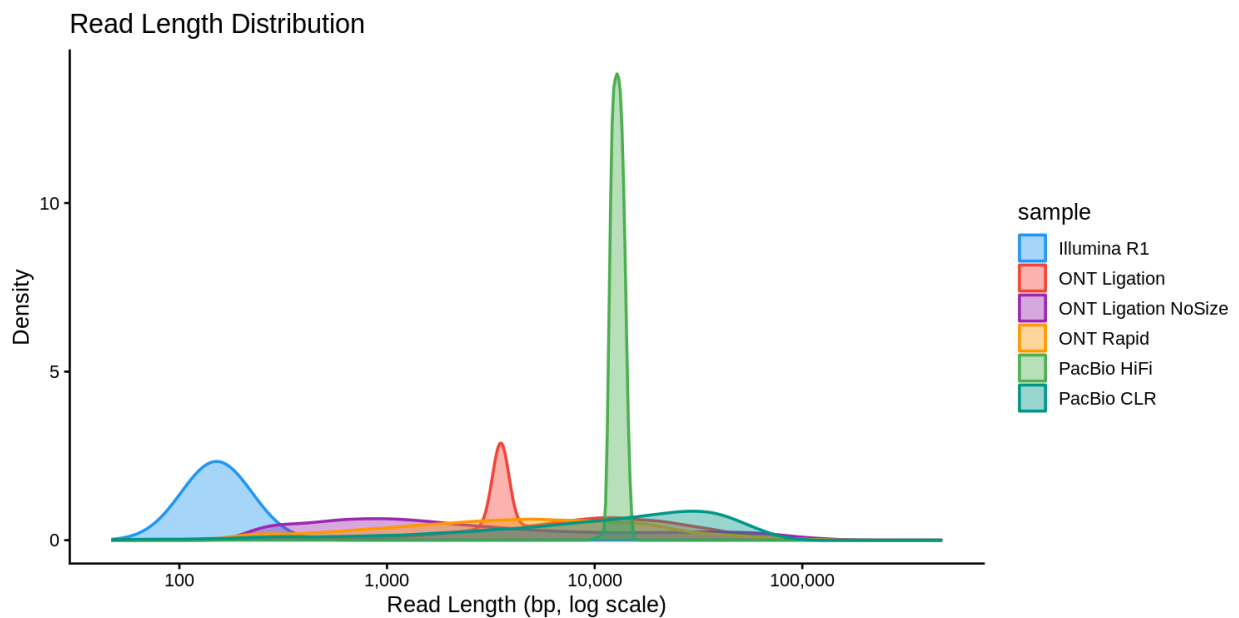
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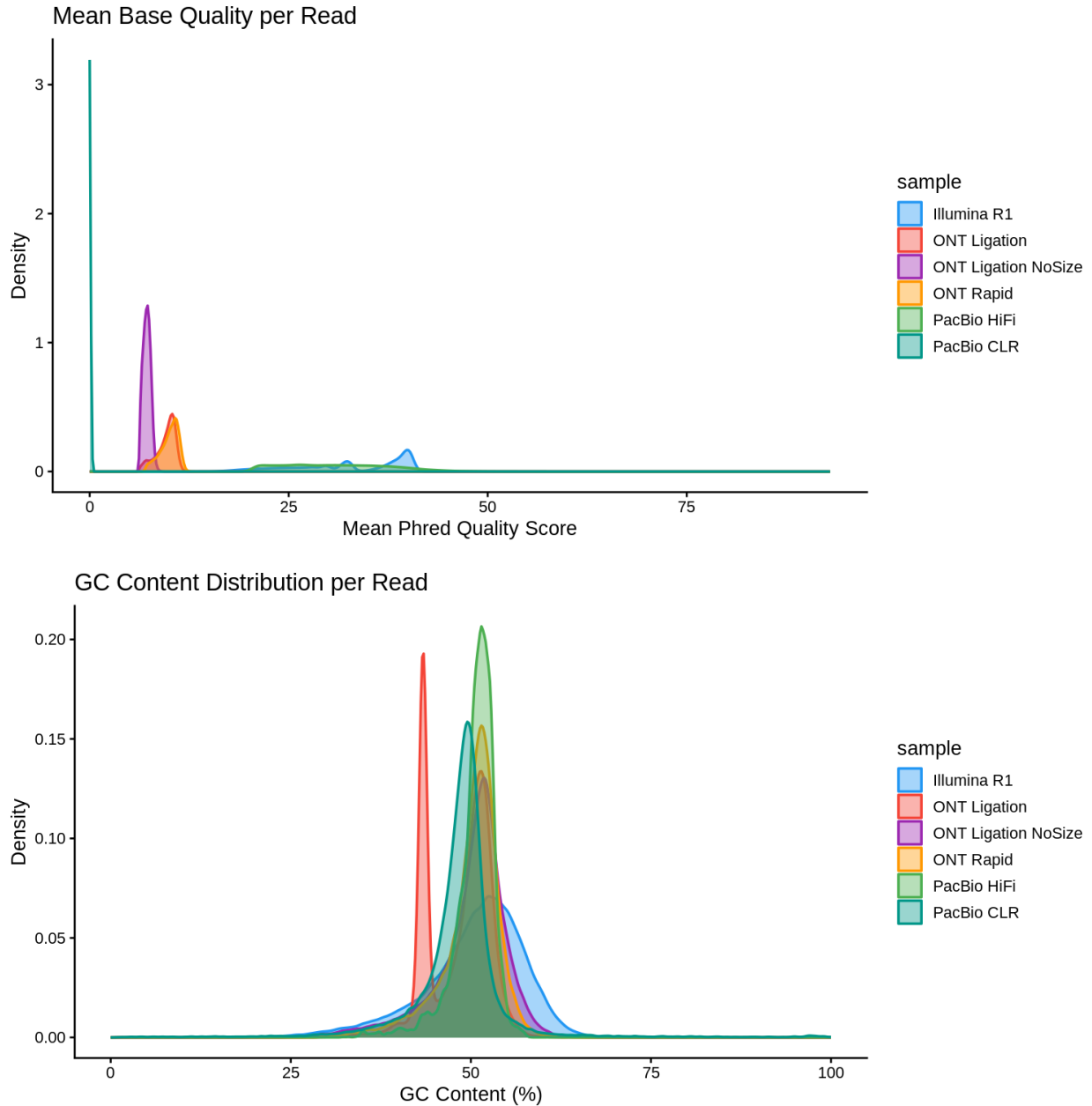
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Task 1: Compare raw read statistics between sequencing technologies and library preparation protocols.

1. Plot a histogram of each of the three metrics. Plot all samples together, each metric in a different plot, for a total of 3 plots (10pts)





2. Is there any difference in read length, base quality and GC content between library preparations for ONT? If so, what is causing these differences? (10pts)

- **Read Length Distribution:** Among the three ONT library preparations, ONT Ligation had the most 137,014 reads, followed by ONT Ligation NoSize 97,301 reads, and ONT Rapid had the fewest at only 31,032 reads. ONT Ligation had the longest median read length 5,962 bp, followed by ONT Rapid 4,254 bp, and ONT Ligation NoSize had the shortest median 1,661 bp. ONT Ligation has longer reads because the ligation includes a size selection step that removes short DNA fragments before sequencing, enriching for longer reads. ONT Ligation NoSize without uses size selection, so short fragments are retained, so the median read length decreases. ONT Rapid uses a transposase enzyme to randomly fragment and ligate adapters to DNA without size selection, resulting in more shorter read lengths compared to ONT Ligation.

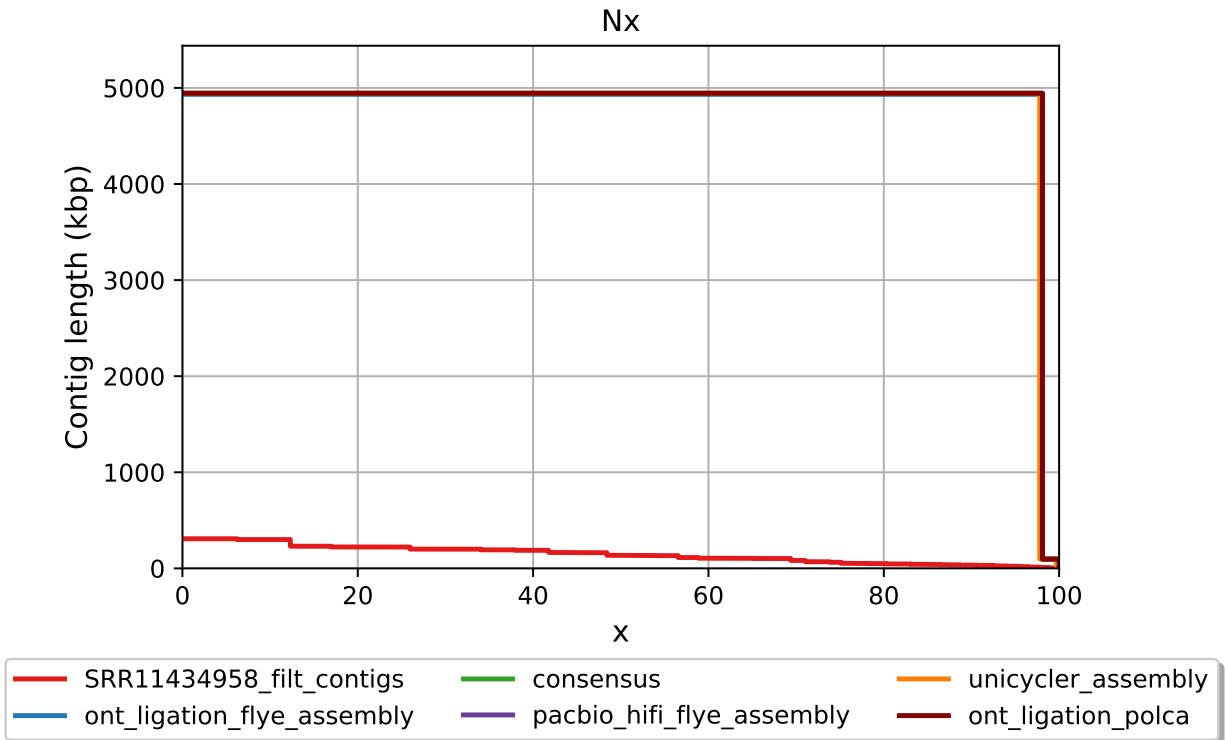
- Base Quality: ONT Rapid had the highest average quality (AvgQual = 10.05), followed by ONT Ligation (AvgQual = 9.4), and ONT Ligation NoSize had the lowest (AvgQual = 7.18). On the plot, ONT ligation and ONT Rapid showed similar base quality distribution. The slightly lower quality in ONT Ligation NoSize could be because shorter reads tend to have lower quality scores, and since NoSize retains many short fragments, this brings the average quality down slightly.
- GC Content: All three ONT library preparations have pretty similar GC content, all centered around 40-50%, which matches the expected GC content of the E. coli genome (~50%). ONT Ligation has a slightly narrower peak at around 40%, while ONT Rapid and ONT NoSize peak around 50%. Overall there aren't really any differences between the three.

3. Is there any difference in read length, base quality and GC content between library preparations for PacBio? If so, what is causing these differences? (10pts)

- Read Length Distribution: PacBio HiFi shows a sharp, narrow peak at around 10,000 bp, while PacBio CLR has a much broader distribution with no clear peak, ranging from about 1,000 bp to over 100,000 bp. HiFi had 89,411 reads and CLR had 63,117 reads. The differences are due to HiFi uses the sequencer reads the same DNA molecule multiple times to correct errors. Therefore, only DNA fragments of a certain size (~10-15 kb) work well, which is why the read lengths are so uniform (Q1=12,286 bp, Q3=13,534 bp). CLR only reads each molecule once, so it can sequence fragments of many different sizes, giving a much wider range of read lengths (Q1=4,369 bp, Q3=29,100 bp) and a higher max length of 122,904 bp.
- Base Quality: PacBio HiFi and CLR look very different on the base quality plot. CLR has a huge spike at around Q0, but this doesn't mean the reads are bad - it's just that CLR uses a different quality score encoding that doesn't work with standard Phred scores, so everything shows up as 0. HiFi on the other hand has a much higher average quality (AvgQual = 27.23), with 98.4% of bases passing Q20 and 96.19% passing Q30. This makes sense because HiFi reads the same DNA molecule multiple times, which corrects most of the errors and gives more accurate base calls compared to CLR.
- GC Content: Both PacBio HiFi and CLR have pretty similar GC content, both peaking around 50%. But HiFi has higher density, meaning its reads have more consistent GC content across all reads. This difference is caused by the size selection in HiFi sequencing.

Task 2: Assessing assembly quality

1. Interpret the Nx plot produced by QUAST. Explain the plot, the axes, what it shows and why is important (5pts)



- The Nx plot shows how contiguous a genome assembly is. X-axis: the x value from 0 to 100, representing the percentage of the total assembly length. Y-axis: contig length in kbp. The Nx plot works by sorting all contigs from longest to shortest. At a given x value, the y value represents the length of the contig at which x% of the total assembly has been accumulated. For example, at N50 (x=50), the y value is the length of the contig at which 50% of the total assembly length has been reached. A higher y value at any given x indicates a more contiguous assembly with fewer, longer contigs.
- In the plot, all the long-read assemblies (ONT Ligation Flye, Medaka, PacBio HiFi, Unicycler, POLCA) are overlapping at around 5,000 kbp and stay consistent almost all the way to x=98-99% before dropping. This means almost the whole genome is assembled into just one or two big contigs. The Illumina assembly stays low at around 100-300 kbp and gradually decreases, showing that it's broken into a lot of smaller contig fragments.
- The Nx plot is important because it gives an overall picture of assembly contiguity. For example, ONT Ligation Flye assembly and PacBio HiFi Flye assembly showed that each have 2 contigs and 0 scaffolds. However, looking at the Nx plot, the second contig drops sharply near x=98-99% to a very short length, which suggests it is likely a plasmid rather than part of the main chromosome. This means ONT Ligation and PacBio HiFi actually only produced 1 true chromosomal contig, which is the ideal result for a bacterial genome like *E. coli*.

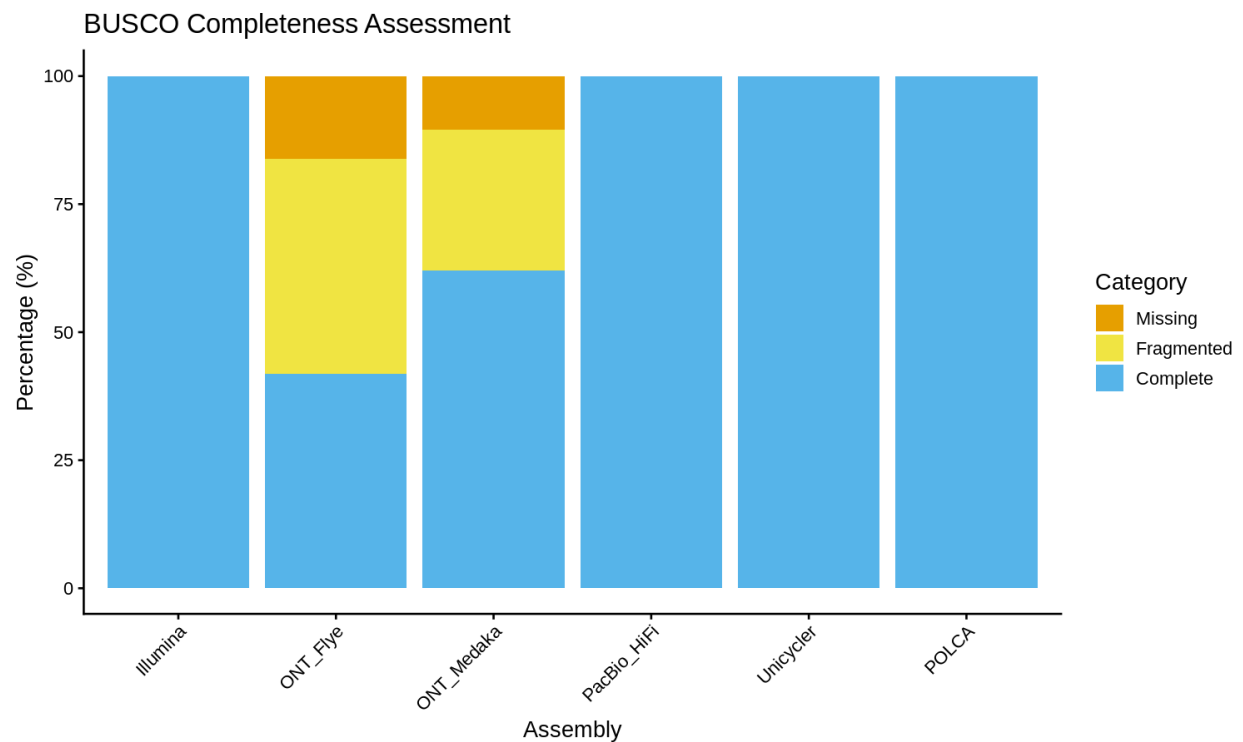
2. What are the sequencing technologies that produce the most contiguous assembly? Does short-read or long-read sequencing produce the most contiguous contigs? (5pts)

- Looking at the Nx plot and QUAST results, all long-read assemblies (ONT Ligation Flye, ONT Medaka, PacBio HiFi, Unicycler, and POLCA) produced the most contiguous assemblies. As discussed above, the second contig for long-read sequencing is likely a plasmid, meaning each of these technologies essentially assembled the entire *E. coli* chromosome into a single contig. In contrast, the Illumina short-read assembly produced 86 contigs.

3. What's the most accurate sequencing technology? (5pts)

- Based on the QUAST report, Illumina had the highest accuracy with 0 misassemblies, only 0.22 mismatches per 100 kbp and 0.16 indels per 100 kbp. This matches with Illumina has very high base quality ($Q30 > 93\%$) and short reads are less likely to cause large-scale misassemblies.
- For long-read technologies, Unicycler and PacBio HiFi performed best with only 1.23 and 1.55 mismatches per 100 kbp, and very low indel rates.
- But ONT Flye had the lowest accuracy with 281 mismatches and 232 indels per 100 kbp, which improved after Medaka polishing (256 mismatches, 120 indels) but was still much higher than PacBio HiFi.
- Overall, Illumina is the most accurate technology, but among long-read technologies, PacBio HiFi and Unicycler produce the most accurate assemblies.

4. Explain the BUSCO gene completeness plot. What's the plot showing? (5pts)



- Based on the BUSCO plot, Illumina, PacBio HiFi, Unicycler, and POLCA all achieved 100% complete BUSCOs, shown as entirely blue bars with no fragmented or missing genes.

- In contrast, ONT Flye had the worst completeness with only 41.9% complete BUSCOs, 41.9% fragmented, and 16.1% missing. After Medaka polishing, the completeness improved to 62.1% complete, with 27.4% fragmented and 10.5% missing. While Medaka polishing clearly helped, the ONT assembly still had a significant proportion of fragmented and missing genes compared to the other technologies.

5. Based on the BUSCO plot, what sequencing technologies produce the most complete assembly? (5pts)

- Illumina, PacBio HiFi, Unicycler, and POLCA all achieved 100% complete BUSCOs.

6. Can an assembly with a bad Nx plot have a good BUSCO analysis result? Explain your answer (10pts)

- Yes, it is possible for an assembly with a bad Nx plot to still have a good BUSCO result. The Nx plot measures contiguity while BUSCO measures completeness, and these are two completely different aspects of assembly quality.
- For examples, illumina assembly has poor Nx plot with $N50 = 136,146$ bp and $L50 = 12$, contigs = 86. However, it still achieved 100% BUSCO completeness. This is because all the gene sequences are still present and fully intact within those smaller fragments.